ECO 362: Financial Investments
Arbitrage Pricing Theory

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Stochastic discount factor

- From the lecture on "Arbitrage", recall the general asset pricing formula.

$$\mathbb{E}[R_i - R_b] = \frac{\mathrm{Cov}(R_i, -M)}{\mathbb{E}[M]} \tag{1}$$

- M is the stochastic discount factor that prices all assets.
- From the lecture on "CAPM",

$$\begin{aligned}\mathbb{E}[R_i - R_b] =& \overline{\gamma}\mathrm{Cov}(R_i, R_m) \\ =& \beta_i \mathbb{E}[R_m - R_b]\end{aligned}$$

- What is the connection?

CAPM as a special case

- CAPM is a special case of the general asset pricing formula when the stochastic discount factor is

$$M = \frac{1 - \overline{\gamma}(R_m - \mathbb{E}[R_m])}{1 + R_b}$$

- Check:

$$\mathbb{E}[M] = \frac{1}{1 + R_b}$$
$$\mathrm{Cov}(R_i, -M) = \frac{\overline{\gamma}\mathrm{Cov}(R_i, R_m)}{1 + R_b}$$

which imply that

$$\mathbb{E}[R_i - R_b] = \overline{\gamma}\mathrm{Cov}(R_i, R_m)$$

Arbitrage Pricing Theory

- CAPM is a one-factor model.
- More generally, a stochastic discount factor could have multiple factors.

$$M = \frac{1 - \sum_{j=1}^{J} b_j (F_j - \mathbb{E}[F_j])}{1 + R_b} \tag{2}$$

- The factors could be
  1. Portfolio returns on various trading strategies.
  2. Macro factors: GDP growth, inflation,...
- We will focus on 1 in this class.
- In the following derivations, we will assume that the factors are uncorrelated with each other.
  - This assumption is unnecessary if we use linear algebra.

Multi-factor model

- Substitute (2) into (1):

$$\mathbb{E}[R_i - R_b] = \sum_{j=1}^{J} b_j \mathrm{Cov}(R_i, F_j) = \sum_{j=1}^{J} b_j \mathrm{Var}(F_j) \underbrace{\frac{\mathrm{Cov}(R_i, F_j)}{\mathrm{Var}(F_j)}}_{\beta_{i,j}}$$

- If F_j is an excess (i.e., long minus short) portfolio return,

$$\mathbb{E}[F_j] = b_j \mathrm{Var}(F_j)$$

- Thus

$$\mathbb{E}[R_i - R_b] = \sum_{j=1}^{J} \beta_{i,j} \mathbb{E}[F_j]$$

Interpreting a multi-factor model

- When you go shopping for fruit,

$$\text{Price} = Q(\text{bananas}) \times P(\text{bananas}) + Q(\text{apples}) \times P(\text{apples})$$

- The price of asset i is

$$\mathbb{E}[R_i - R_b] = \sum_{j=1}^{J} \beta_{i,j} \mathbb{E}[F_j]$$

  - $\beta_{i,j}$ is the quantity of risk factor j that asset i has.
  - $\mathbb{E}[F_j]$ is the price of risk factor j.
- Recall that only systematic risk is priced (i.e., $\mathbb{E}[F_j] = 0$ for idiosyncratic risk).

Fama-French 3-factor model

	ME below 50%	ME above 50%
BE/ME above 70%	Small Value	Big Value
BE/ME 30–70%	Small Neutral	Big Neutral
BE/ME below 30%	Small Growth	Big Growth

1. Market factor: Market minus riskless.
2. Size factor: Small minus Big (SMB) market equity.

$$\begin{aligned} \text{SMB} = & \frac{1}{3}(\text{Small Value} + \text{Small Neutral} + \text{Small Growth}) \\ & - \frac{1}{3}(\text{Big Value} + \text{Big Neutral} + \text{Big Growth}) \end{aligned}$$

3. Value factor: High minus Low (HML) book-to-market equity.

$$\begin{aligned} \text{HML} = & \frac{1}{2}(\text{Small Value} + \text{Big Value}) \\ & - \frac{1}{2}(\text{Small Growth} + \text{Big Growth}) \end{aligned}$$

Interpreting the 3-factor model

- According to the Fama-French 3-factor model, expected return on asset i is

$$\mathbb{E}[R_i - R_b] = \beta_{i,m}\mathbb{E}[R_m - R_b] + \beta_{i,SMB}\mathbb{E}[\text{SMB}] + \beta_{i,HML}\mathbb{E}[\text{HML}]$$

1. Market beta.
2. SMB beta.
   - $\beta_{i,SMB} > 0$: Portfolio tilts toward small-cap stocks.
   - $\beta_{i,SMB} < 0$: Portfolio tilts toward large-cap stocks.
3. HML beta.
   - $\beta_{i,HML} > 0$: Portfolio tilts toward value stocks.
   - $\beta_{i,HML} < 0$: Portfolio tilts toward growth stocks.

Estimating the 3-factor model

- Using time-series data on returns, estimate a linear regression for each asset i.

$$R_{i,t} - R_{b,t} = \alpha_i + \beta_{i,m}(R_{m,t} - R_{b,t}) + \beta_{i,SMB}\mathrm{SMB}_t + \beta_{i,HML}\mathrm{HML}_t + \epsilon_{i,t}$$

- α_i is alpha.
  - Test whether $\alpha_i = 0$ for all assets.
  - If $\alpha_i > 0$, average return is too high relative to risk.
- The residual $\epsilon_{i,t}$ represents idiosyncratic risk that is uncorrelated with the factors.
- Data on factors and portfolio returns available at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/

Constructing portfolios sorted by market equity

1. Sort stocks into 10 portfolios based on market equity (i.e., cutoff at each 10th percentile).
2. Compute returns on these portfolios over the subsequent year.
3. Rebalance annually.

Example 1: Portfolios sorted by market equity

Statistic	Low	2	3	4	5	6	7	8	9	High
	Portfolio									
Mean (%)	1.09	0.98	0.96	0.93	0.88	0.89	0.82	0.81	0.74	0.63
SD (%)	9.81	8.69	7.91	7.43	7.02	6.78	6.41	6.11	5.78	5.05
Sharpe ratio	0.11	0.11	0.12	0.12	0.13	0.13	0.13	0.13	0.13	0.12
Market beta	1.00	1.06	1.07	1.05	1.06	1.07	1.05	1.05	1.03	0.98
SMB beta	1.55	1.28	1.01	0.88	0.73	0.50	0.41	0.23	0.07	−0.21
HML beta	0.75	0.48	0.37	0.31	0.20	0.23	0.14	0.12	0.11	−0.04
Alpha (%)	−0.14	−0.14	−0.08	−0.05	−0.04	−0.01	−0.01	0.02	0.00	0.03
Alpha (t-stat)	−1.46	−2.74	−2.25	−1.53	−1.22	−0.20	−0.25	0.44	−0.12	1.79

Interpreting the 3-factor regression

- Portfolio 1.

$$R_{1,t} - R_{b,t} = -0.14\% + 1.00(R_{m,t} - R_{b,t}) + 1.55 \times \mathrm{SMB}_t + 0.75 \times \mathrm{HML}_t + \epsilon_{1,t}$$

- Portfolio strategy:
  - Long \$1 in market.
  - 1.55×SMB: Long \$1.55 in Small and short \$1.55 in Big.
  - 0.75×HML: Long \$0.75 in High and short \$0.75 in Low.

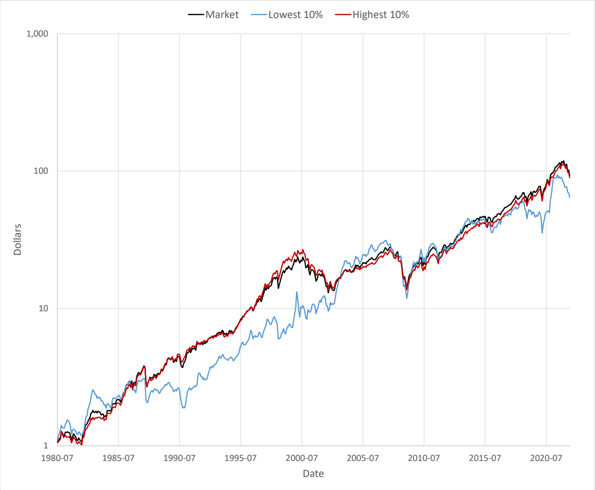
- Portfolio 10.

$$R_{10,t} - R_{b,t} = 0.03\% + 0.98(R_{m,t} - R_{b,t}) - 0.21 \times \text{SMB}_t - 0.04 \times \text{HML}_t + \epsilon_{10,t}$$

- Portfolio strategy:
  - Long \$0.98 in market and long \$0.02 in riskless.
  - $-0.21\times$SMB: Short \$0.21 in Small and long \$0.21 in Big.
  - $-0.04\times$HML: Short \$0.04 in High and long \$0.04 in Low.

Cumulative returns on low vs. high market equity

Cumulative returns on low vs. high market equity (1980–2022)



Constructing portfolios sorted by book-to-market equity

1. Sort stocks into 10 portfolios based on book-to-market equity (i.e., cutoff at each 10th percentile).
2. Compute returns on these portfolios over the subsequent year.
3. Rebalance annually.

Example 2: Portfolios sorted by book-to-market equity

	Portfolio									
Statistic	Low	2	3	4	5	6	7	8	9	High
Mean (%)	0.62	0.72	0.69	0.66	0.73	0.80	0.70	0.92	1.05	1.05
SD (%)	5.68	5.30	5.39	5.90	5.60	6.03	6.39	6.74	7.63	9.12
Sharpe ratio	0.11	0.14	0.13	0.11	0.13	0.13	0.11	0.14	0.14	0.12
Market beta	1.08	0.98	0.98	1.03	0.96	0.98	1.00	1.02	1.09	1.18
SMB beta	−0.08	−0.01	−0.04	−0.02	−0.06	−0.05	0.03	0.12	0.24	0.55
HML beta	−0.33	−0.19	−0.04	0.15	0.25	0.43	0.55	0.67	0.84	1.08
Alpha (%)	0.03	0.14	0.05	−0.07	0.00	0.00	−0.18	−0.02	−0.02	−0.23
Alpha (t-stat)	0.72	3.30	1.21	−1.38	0.08	0.08	−3.35	−0.48	−0.45	−3.17

Interpreting the 3-factor regression

- Portfolio 1.

$$R_{1,t} - R_{b,t} = 0.03\% + 1.08(R_{m,t} - R_{b,t}) - 0.08 \times \mathrm{SMB}_t \\ - 0.33 \times \mathrm{HML}_t + \epsilon_{1,t}$$

- Portfolio strategy:
  - Long \$1.08 in market and short \$0.08 in riskless.
  - $-0.08\times$SMB: Short \$0.08 in Small and long \$0.08 in Big.
  - $-0.33\times$HML: Short \$0.33 in High and long \$0.33 in Low.

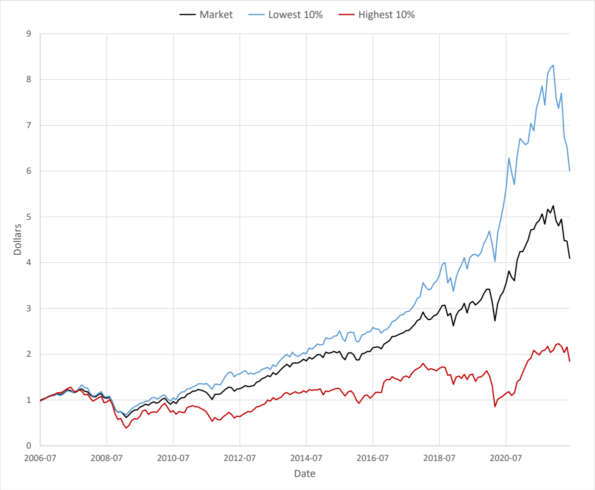
- Portfolio 10.

$$R_{10,t} - R_{b,t} = -0.23\% + 1.18(R_{m,t} - R_{b,t}) + 0.55 \times \text{SMB}_t + 1.08 \times \text{HML}_t + \epsilon_{10,t}$$

- Portfolio strategy:
  - Long \$1.18 in market and short \$0.18 in riskless.
  - $0.55\times$SMB: Long \$0.55 in Small and short \$0.55 in Big.
  - $1.08\times$HML: Long \$1.08 in High and short \$1.08 in Low.

Cumulative returns on low vs. high book-to-market equity

Cumulative returns on low vs. high book-to-market equity (2006–2022)



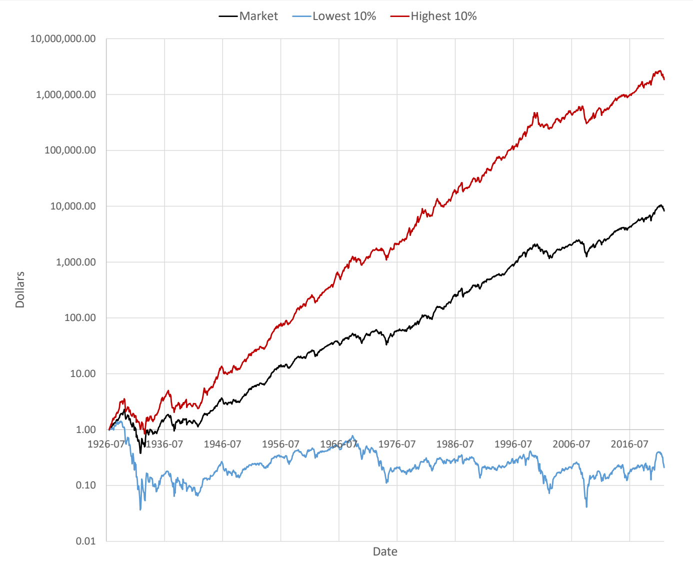
Constructing portfolios sorted by prior returns

1. Sort stocks into 10 portfolios based on prior annual return (i.e., cutoff at each 10th percentile).
2. Compute returns on these portfolios over the subsequent month.
3. Rebalance monthly.

Example 3: Portfolios sorted by prior returns

	Portfolio									
Statistic	Low	2	3	4	5	6	7	8	9	High
Mean (%)	0.04	0.45	0.47	0.62	0.62	0.69	0.74	0.87	0.93	1.25
SD (%)	9.73	8.05	6.95	6.31	5.93	5.78	5.47	5.31	5.61	6.43
Sharpe ratio	0.00	0.06	0.07	0.10	0.10	0.12	0.14	0.16	0.17	0.19
Market beta	1.39	1.25	1.12	1.06	1.01	1.02	0.98	0.95	0.98	1.02
SMB beta	0.50	0.13	−0.02	−0.05	−0.06	−0.04	−0.11	−0.07	−0.02	0.29
HML beta	0.44	0.39	0.34	0.25	0.24	0.16	0.07	−0.01	−0.04	−0.31
Alpha (%)	−1.15	−0.55	−0.40	−0.17	−0.13	−0.04	0.09	0.26	0.30	0.63
Alpha (*t*-stat)	−8.25	−5.24	−4.74	−2.52	−2.21	−0.77	1.65	4.62	4.60	6.54

Cumulative returns on low vs. high prior returns



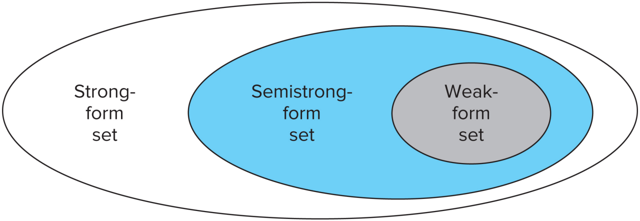
Summary of anomalies

1. Low beta: Low-beta stocks outperform high-beta stocks (relative to CAPM).
2. Low volatility: Low-volatility stocks outperform high-volatility stocks.
3. Value: High book-to-market equity stocks outperform low book-to-market equity stocks.
4. Momentum: Winner stocks (i.e., high prior annual return) outperform loser stocks.

- These anomalies also exist in international equity markets.
- Tradeable strategies as ETFs.
- Warning: Past performance does not necessarily predict future results.

Efficient-market hypothesis

1. Weak form: Asset prices reflect information that can be inferred from historical prices and returns.
2. Semi-strong form: Asset prices reflect all publicly available information.
3. Strong form: Asset prices reflect even private or "insider" information.



Source: Bodie, Kane, and Marcus (2018, Chapter 11)

Test 1: Reaction to earnings surprises

- Firms are required to disclose earnings every quarter.
- Test the semi-strong form of market efficiency.
  - Only earnings surprises should matter.
  - Stock price should immediately and fully react upon announcement.
- Analyst forecast from Institutional Brokers Estimate System.
- Estimate surprises as difference between actual earnings and analyst forecast.
- Classify a large sample of earnings surprises into 10 deciles: 1 (most negative) to 10 (most positive).

Event study methodology

1. Estimate CAPM on daily returns prior to the event date (assuming constant riskless rate).

$$R_{i,t} = \alpha_i + \beta_{i,m} R_{m,t} + \epsilon_{i,t}$$

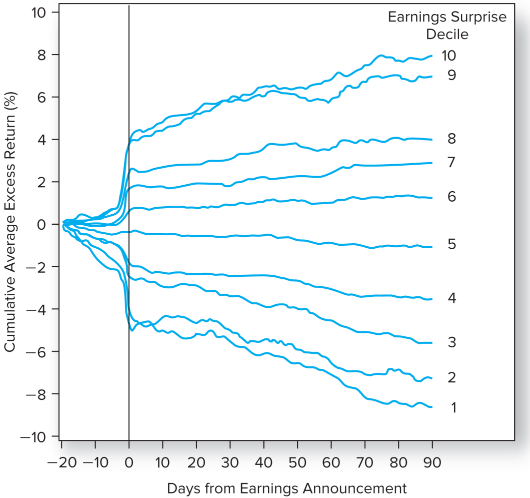
2. Use estimates $\widehat{\alpha}_i$ and $\widehat{\beta}_{i,m}$ to compute abnormal returns for the whole sample.

$$\mathrm{AR}_{i,t} = R_{i,t} - \widehat{\alpha}_i - \widehat{\beta}_{i,m} R_{m,t}$$

3. Compute cumulative abnormal returns.

- Abnormal return isolates the impact of the event in the absence of market fluctuations.
- Alternatively, estimate 3-factor model instead of the CAPM.

Cumulative abnormal return around earnings surprises



Source: Bodie, Kane, and Marcus (2018, Figure 11.5)

Rejection of efficient-market hypothesis?

- Price drift before announcement could be explained by
  - Soft information that is publicly available ahead of announcement.
  - Inside information?
- Price drift after announcement is hard to explain if market is efficient. Delayed reaction to news suggests
  - Inattentive investors that do not pay attention.
  - Overreaction to news because of momentum traders.

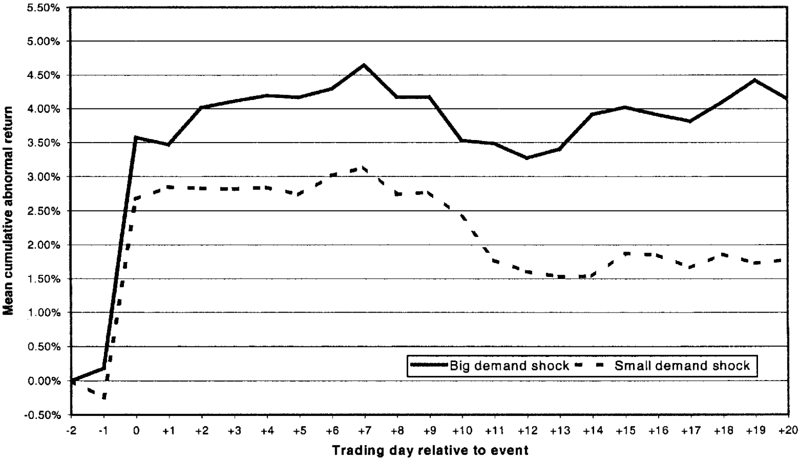
Test 2: Reaction to index additions

- Firms get added to or deleted from the S&P 500 index.
- Decided by a committee based on market cap, liquidity, and other criteria.
- Index additions are preannounced.
- Test the semi-strong form of market efficiency.
  - Addition reflects past good news, which should already be incorporated in stock price.

Data

- Sample of 278 index additions between between September 22, 1976 and September 30, 1989.
- Daily returns for 20 days before and after the index addition.
- Classify events based on size of S&P 500 index funds relative to the stock's market cap.
    - Big demand shock: Smaller cap stocks.
    - Small demand shock: Larger cap stocks.

Cumulative abnormal return around index additions



Source: Wurgler and Zhuravskaya (2002, *Journal of Business*)

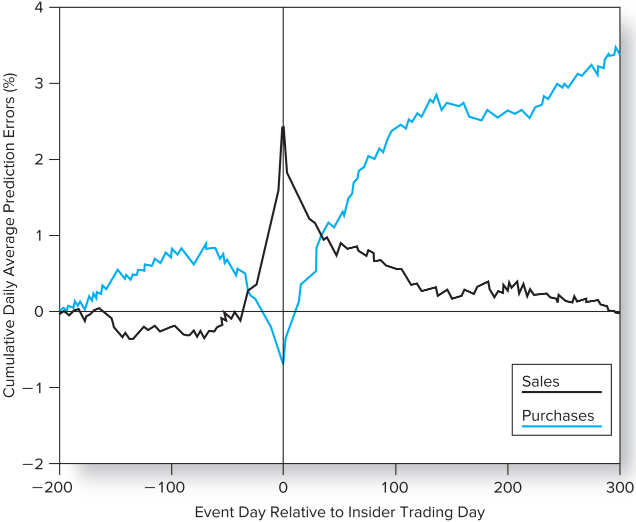
Rejection of efficient-market hypothesis?

- S&P 500 index funds are required to hold stocks in the index.
- There may be nearly identical stocks (in terms of risk) that are outside the index that these funds cannot hold.
- This causes stocks in the S&P 500 index to be overpriced.
- There are not enough arbitrageurs to fix this mispricing.

Test 3: Reaction to insider trades

- Insiders: Officers, directors, and large shareholders (greater than 10%).
- Must file their trades within 2 days on SEC Form 4 (since the Sarbanes-Oxley Act of 2002).
- Looser filing deadline prior to 2002.
- Data on insider trades between 1975 to 1981.
- 66% of trades filed within 30 days, 27% in 30–60 days, and 7% in greater than 60 days.

Cumulative abnormal return around insider trades



Source: Bodie, Kane, and Marcus (2018, Chapter 11)

Rejection of efficient-market hypothesis?

- Evidence shows that insiders time their sales and purchases.
  - Sell when price is expected to fall.
  - Buy when price is expected to rise.
- Rejection of the strong-form of market efficiency.
- Perhaps not surprising.

Other uses of event studies

- Estimating the market value of new products.
  - FDA approvals of new drugs.
- Estimating damages.
  - Environmental disasters like oil spills.
  - Product recalls.
- Estimating the impact of government policy.
  - Trade policy.
  - Anti-trust regulation.
- Great tool for a fun senior thesis topic!

Summary

- Multi-factor models to evaluate the performance of trading strategies.
- Event studies to test efficient-market hypothesis and to estimate the market value of new products, damages, and government policy.